

AREDN Mesh Metrics Dashboard

aredn_metrics_dashboard.html — User Documentation

Version 1.8.0 | William Telford (WB3CQK)

Developed with assistance from Anthropic Claude AI

1. Overview

aredn_metrics_dashboard.html is a self-contained, single-file web dashboard for monitoring AREDN (Amateur Radio Emergency Data Network) mesh nodes in real time. It requires no installation, no web server, and no internet connection — open it in any modern browser and it works immediately.

The dashboard polls your AREDN nodes directly from the browser using the sysinfo JSON API, displaying node health, RF link quality, antenna aiming metrics, GPS location, and alerts.

2. Technology

2.1 Languages Used

Language	Role
HTML5	Page structure, form controls, tab panels, and table layouts.
CSS3	Styling, dark mode, responsive layout, CSS custom properties for theming, and keyframe animations.
JavaScript (ES2020)	All application logic — polling, parsing, rendering, alerts, history, aiming, discovery, export, and map display.

2.2 Key Browser APIs Used

- `fetch()` — HTTP requests to AREDN node sysinfo API endpoints.
- `AbortController` — enforces a 10-second timeout on each fetch request.
- `Promise.all()` — polls all nodes and fetches all neighbour sysinfo in parallel.
- `AudioContext` — generates alert beep tones without any audio files.
- `URL.createObjectURL()` — enables the JSON export download without a server.
- `navigator.userAgent` — detects Android to display the IP-address tip.
- `prefers-color-scheme` — automatic dark mode support.

2.3 API Endpoints

The dashboard queries the following AREDN endpoints:

- `/a/sysinfo?lqm=1` — primary endpoint, AREDN firmware 3.20 and later. Returns node health, LQM link data, RF configuration, GPS, and tunnel counts.
- `/cgi-bin/sysinfo.json?lqm=1` — legacy fallback for older firmware. Automatically tried if the primary endpoint returns HTTP 404.
- `/a/sysinfo?hosts=1` — used by the Discover feature to fetch the full host list from a seed node for auto-discovery.

- `/cgi-bin/metrics` — Prometheus-style endpoint fetched to retrieve the RF noise floor value (firmware 3.23+).

Neighbour firmware and routing protocol are fetched in parallel from each visible neighbour's sysinfo endpoint after the main poll completes.

2.4 No External Dependencies

The file is entirely self-contained. There are no CDN requests, no npm packages, no external fonts, and no internet connection required at runtime.

3. Getting Started

3.1 Opening the Dashboard

Double-click `aredn_metrics_dashboard.html` or open it in a browser via File > Open. The configuration panel is shown immediately.

3.2 Connecting to Nodes

Enter one or more node IPs or names and click Start polling.

Field	Description
Node IP or name	Enter the management IP (e.g. 10.1.2.3) or node name (e.g. WB3CQK-NODE01). A plain name without dots automatically has <code>.local.mesh</code> appended.
Refresh every	Poll interval in seconds. Default is 5 seconds. Minimum is 1 second. Use 1–2 seconds for antenna aiming; 5–30 seconds for general monitoring.

Click + Add node to add more nodes. Click Stop to pause polling.

3.3 Auto-Discovery

Click ⊕ Discover in the topbar to query the first node's host list and automatically populate the node list with all discovered mesh nodes. Discovery filters out known non-node hosts (phones, cameras, VoIP adapters, etc.).

3.4 Android

On Android, use the node IP address rather than the node name. Android devices may not resolve `.local.mesh` hostnames. Android Chrome may block fetch requests from files opened directly from Downloads. Serve the dashboard from a node's web server as described in section 3.5.

3.5 Deploying to a Node Web Server

Copy the HTML file to an AREDN node's `/www` directory and open it over HTTP. This avoids all browser security restrictions and is recommended for Android.

```
scp -O -P 2222 aredn_metrics_dashboard.html root@10.x.x.x:/www/
```

Once copied, open `http://10.x.x.x/aredn_metrics_dashboard.html` in any browser on the mesh network.

4. Interface Overview

4.1 Header Bar

- ⊕ Discover — auto-discovers all mesh nodes from the first seed node.
- ↓ Export — downloads the current poll data as a timestamped JSON file.

4.2 Summary Cards

- Nodes online — count of configured nodes that responded on the last poll.
- Active links — total LQM neighbour links visible across all polled nodes.
- Avg quality — mean LQM quality percentage across all visible links.
- Best link — hostname of the single highest-quality neighbour link.

4.3 Tabs

- Nodes — health card per node: hardware, firmware, RF config, load average, and LQM links.
- RF Links — unified sortable table of all LQM links.
- Antenna Aiming — large-format signal display for real-time antenna adjustment.

5. Nodes Tab

5.1 Node Health Fields

Field	Description
Model	Hardware model string reported by the node.
Firmware	Installed AREDN firmware version string.
Routing	Detected routing protocol: Babel (green), OLSR (amber), or Babel+OLSR (blue).
Uptime	System uptime as reported by the node.
Free RAM	Free memory converted from KB to MB for display.
Tunnels	Number of active tunnel connections.
Location	GPS latitude and longitude in decimal degrees, if configured on the node.
API version	sysinfo API version string (1.x = OLSR era, 2.x = Babel era).

5.2 Load Average Bars

Three bars show the Linux load average at 1-minute, 5-minute, and 15-minute intervals. Green = below 0.8, Amber = 0.8 to 1.5, Red = above 1.5.

5.3 Neighbour Link Table

Each node card shows a compact table of its visible LQM neighbours with hostname, IP, quality, ping, signal, SNR (when available), routing protocol, and firmware version.

6. RF Links Tab

The RF Links tab shows a unified table of every LQM link across all polled nodes, sortable by Quality, Signal, Ping, or SNR.

6.1 Columns

Column	Description
Hostname	Neighbour hostname with source node and mesh IP below.
Quality	LQM composite quality percentage with colour bar.
Ping	Round-trip latency in milliseconds.
Signal	Received signal level in dBm.
SNR local	Local signal-to-noise ratio in dB.
SNR remote	Reverse SNR as reported by the neighbour.
Babel metric	Babel routing cost. Lower = better path. — on OLSR nodes.
Routing	Routing protocol on the neighbour node.
Firmware	Firmware version of the neighbour.

7. Understanding Link Metrics

7.1 Link Quality

Link Quality is AREDN's LQM composite score (0–100%). It reflects packet delivery success, ping round-trip success rate, and link stability. It does not directly measure signal strength or RF headroom.

7.2 Signal Level (dBm)

Received RF power at the antenna. Typical values range from –40 dBm (strong) to –85 dBm (marginal). Signal alone does not determine link quality — noise floor matters equally.

7.3 SNR — Signal-to-Noise Ratio (dB)

The difference between received signal level and the local noise floor. Good: 30 dB+. Marginal: 15–25 dB. Poor: below 15 dB. SNR remote is the SNR measured at the far end of the link.

7.4 Babel Routing Metric

A cost value assigned to each link by the Babel routing protocol. Lower values indicate a better path. A falling Babel metric during antenna aiming confirms the routing layer has recognised the improvement.

8. Antenna Aiming Tab

Large-format real-time signal display for use while physically adjusting an antenna. Updates on every poll cycle.

8.1 Display Elements

Element	Description
Link Quality bar	Quality % with large visual bar and delta vs session best.
Ping Latency	Round-trip time in ms with delta vs session best.
Signal (RX)	Received signal in dBm. Higher (less negative) is better.
SNR (local)	Local SNR in dB. Higher is better.
SNR (remote)	Reverse SNR at the far end of the link.
Babel Metric	Babel routing cost. Lower is better.
Best seen bar	Peak values for each metric recorded in the current session.

8.2 Aiming Workflow

- Start polling the local node. Navigate to the Antenna Aiming tab.
- Select the target neighbour. Click Lock target to freeze the selection.
- Adjust the antenna — watch SNR local and SNR remote rise together.
- Quality reaching 100% does not mean aiming is complete. Maximise SNR.
- A falling Babel metric confirms the routing table has recognised the improvement.

9. Alerts

Setting	Description
Low quality alert	Triggers when any link quality drops below the configured threshold (default 70%).
Node offline alert	Triggers when a node that was previously online stops responding.
Play a sound	Generates a short audio beep via AudioContext when an alert fires.

10. Export

The ↓ Export button downloads a JSON file of the current poll data named `aredn_metrics_YYYY-MM-DDTHH-MM-SS.json`. The file contains a JSON array of node objects with health and link fields matching the field names listed in section 11.

11. Source Metrics Reference

This section lists every data field read by the dashboard's JavaScript. Fields are shown using their exact JSON key names as they appear in the AREDN API response. The dashboard reads no other fields from the API.

11.1 Node Health Fields

Source: `/a/sysinfo` (or `/cgi-bin/sysinfo.json` on older firmware)

These fields are read from the health object inside each polled node's sysinfo JSON response.

11.1.1 Top-level sysinfo fields

JSON Field	Dashboard Label	Notes
sysinfo.uptime	Uptime	String, e.g. '14 days, 3:22:11'
sysinfo.loads[0]	Load 1m	1-minute load average (float)
sysinfo.loads[1]	Load 5m	5-minute load average (float)
sysinfo.loads[2]	Load 15m	15-minute load average (float)
sysinfo.freememory	Free RAM	Integer KB; divided by 1024 to display as MB

11.1.2 node_details block

JSON Field	Dashboard Label	Notes
node_details.model	Model	Hardware model string
node_details.firmware_version	Firmware	Used for display and routing protocol detection

11.1.3 Top-level sysinfo response fields

JSON Field	Dashboard Label	Notes
api_version	API version	1.x = OLSR era, 2.x = Babel era
lat	Location	GPS latitude (float, decimal degrees)
lon	Location	GPS longitude (float, decimal degrees)
tunnels.active_tunnel_count	Tunnels	Integer count of active tunnel connections

routing_protocol is derived by the dashboard from firmware_version and api_version. It is not a raw API field — it is computed locally and not fetched directly from the node.

11.2 LQM Link Fields

Source: `/a/sysinfo?lqm=1` → `lqm.info.trackers` (keyed by neighbour MAC address)

Each entry in the trackers object represents one visible LQM neighbour. The dashboard reads the following fields from each tracker entry:

JSON Field	Dashboard Label	Notes
hostname	Hostname	Neighbour node name
ip	IP	Neighbour mesh IP address (10.x.x.x)
quality	Quality	LQM composite quality, 0–100 (float)
ping_success_time	Ping	Round-trip time in seconds; multiplied by 1000 to display as ms
signal	Signal	Received signal level in dBm (integer); null on older firmware

snr	SNR local	Local signal-to-noise ratio in dB (integer); null on older firmware
rev_snr	SNR remote	Reverse SNR reported by the neighbour (integer); null on older firmware
babel_metric	Babel metric	Babel routing cost (integer); null on OLSR nodes
routable	(internal)	Boolean; used to filter non-routable entries
localarea	(internal)	Boolean; true = RF link, false = tunnel link

11.3 Neighbour Sysinfo Fields

Source: `/a/sysinfo` fetched individually from each neighbour's IP after the main poll

After the main poll the dashboard fetches sysinfo from each visible neighbour in parallel to populate the Routing and Firmware columns. Only two fields are read from each neighbour response:

JSON Field	Dashboard Label	Notes
<code>node_details.firmware_version</code>	Firmware	Neighbour firmware version string; shown as — on fetch error
<code>api_version</code>	Routing	Used together with <code>firmware_version</code> to detect Babel vs OLSR

neighbor_routing and *neighbor_firmware* are the dashboard's own derived fields in the link object — they are not raw API fields. *neighbor_error* holds an error string (e.g. 'timed out') when the neighbour cannot be reached.

11.4 Prometheus Metrics Fields

Source: `/cgi-bin/metrics` (Prometheus text format, firmware 3.23 and later only)

The dashboard fetches this endpoint once per poll cycle per node to retrieve the RF noise floor. The response is in Prometheus text exposition format (one metric per line). The dashboard reads only the following metric name:

Prometheus Metric Name	Dashboard Label	Notes
<code>aredn_rf_noise_dbm</code>	Noise floor	RF noise floor in dBm (float). Shown in the node health card. Not available on firmware older than 3.23 — endpoint returns 404.

Other Prometheus metrics present in the endpoint response (`aredn_rf_signal_dbm`, `aredn_rf_snr_db`, `aredn_rf_tx_mcs`, `aredn_rf_tx_rate_mbps`, `aredn_rf_channel`) are not currently read by the dashboard JavaScript. Only `aredn_rf_noise_dbm` is consumed.

11.5 Complete Field Inventory

The table below lists every distinct API field read by the dashboard JavaScript, across all endpoints.

JSON Field / Prometheus Metric	Endpoint
<code>sysinfo.uptime</code>	<code>/a/sysinfo</code>
<code>sysinfo.loads[0..2]</code>	<code>/a/sysinfo</code>

sysinfo.freememory	/a/sysinfo
node_details.model	/a/sysinfo
node_details.firmware_version	/a/sysinfo and neighbour /a/sysinfo
api_version	/a/sysinfo and neighbour /a/sysinfo
lat	/a/sysinfo
lon	/a/sysinfo
tunnels.active_tunnel_count	/a/sysinfo
lqm.info.trackers[*].hostname	/a/sysinfo?lqm=1
lqm.info.trackers[*].ip	/a/sysinfo?lqm=1
lqm.info.trackers[*].quality	/a/sysinfo?lqm=1
lqm.info.trackers[*].ping_success_time	/a/sysinfo?lqm=1
lqm.info.trackers[*].signal	/a/sysinfo?lqm=1
lqm.info.trackers[*].snr	/a/sysinfo?lqm=1
lqm.info.trackers[*].rev_snr	/a/sysinfo?lqm=1
lqm.info.trackers[*].babel_metric	/a/sysinfo?lqm=1
lqm.info.trackers[*].routable	/a/sysinfo?lqm=1
lqm.info.trackers[*].localarea	/a/sysinfo?lqm=1
aredn_rf_noise_dbm	/cgi-bin/metrics

Revision History

Version / Date	Changes
1.8.0 — 2026-04-20	Added Section 11: Source Metrics Reference — complete mapping of every JSON field and Prometheus metric read by the dashboard JavaScript, with source endpoint, display label, and notes.
1.7.0 — 2026-04-18	Minimum poll interval reduced from 5 seconds to 1 second. Poll timing bar warns when interval is too aggressive for the number of nodes polled.
1.6.0 — 2026-04-18	Added dark/light mode toggle. Added XHR fallback for Android compatibility. Added node web server deployment instructions.
1.5.0 — 2026-04-17	Added noise floor display from /cgi-bin/metrics. Added RF conflict detection alert banner. Dashboard title updated.
1.4.0 — 2026-04-17	Added Topology tab, Compare tab, link detail chart, and poll timing indicator.
1.3.0 — 2026-04-17	Fixed node discovery. Discovered nodes show hostname above IP. Node hostnames used as display labels.
1.2.0 — 2026-04-17	Added alert system, sparklines, antenna aiming target lock, auto-discovery, RF channel/SSID/antenna fields, Map tab, and JSON export.
1.1.0 — 2026-04-17	Removed demo mode. Default refresh 5s. Node entry accepts name or IP. Neighbour firmware/routing fetched in parallel.

1.0.0 — 2026-04-16	Initial release.
--------------------	------------------